Netaji Subhash Engineering College

PRESENTATION ON :- ***Research & Research Objectives:-***

Subject :- Research Methodology

Subject Code :- CS601(6880)

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Class roll no. :- 117

Department :- CSE

Research Methodology: Understanding Research & Research Objectives

A foundational guide to systematic investigation and goal setting.

❑ What is Research?

Research is a systematic investigation undertaken to establish facts, uncover new conclusions, or expand existing knowledge. It is a rigorous process that goes beyond mere information gathering.

It delves deep into unanswered questions, aiming to provide comprehensive and evidence-based answers. This methodical approach forms the backbone of evidence-based decisions and drives innovation across all fields.

❑ What Are Research Objectives?

Clear Goals

Research objectives are precise, concise statements outlining the specific goals your study aims to achieve.

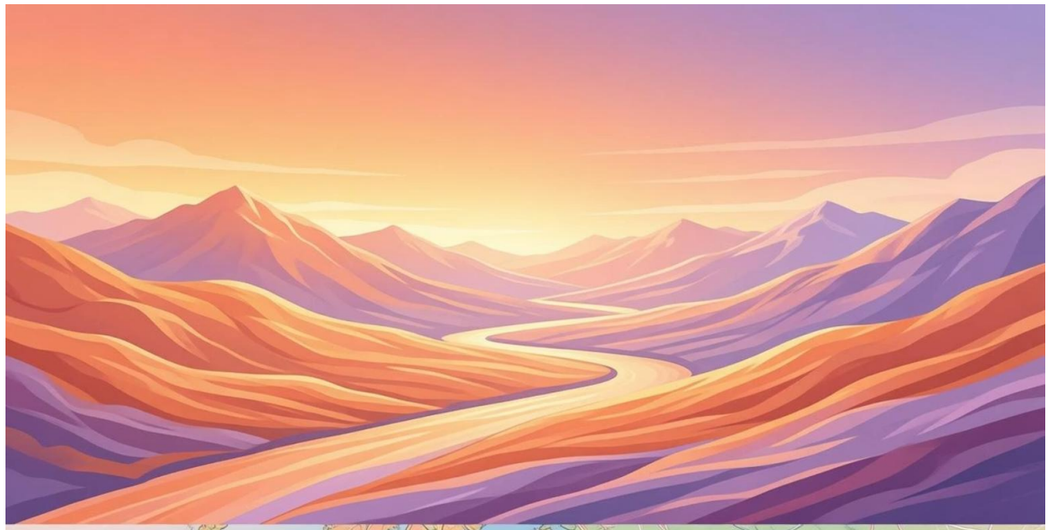
Guiding Principles

They serve as a compass, guiding every step of your research, from data collection and analysis to drawing conclusions.

Defining Scope

Objectives define the exact scope and focus of your study, ensuring efficiency and preventing unnecessary work.

❑ Research Aim vs. Research Objectives



Research Aim:

The broad, overarching purpose of your study. Typically, there is only **one main aim**.

Research Objectives:

The specific, actionable steps required to achieve your research aim. There are usually **multiple objectives**.

Example:

Aim: To examine factors affecting muscle retention in elderly people.

Objectives:

- Assess sedentary habits' impact on muscle atrophy.
- Determine dietary protein's effect on muscle health.
- Evaluate physical activity's role in muscular health.

❑ Why Are Research Objectives Important?

→ Define Scope & Focus

They clearly establish the boundaries and depth of your study, keeping it focused and manageable.

→ Guide Methodology

Objectives directly inform the selection of appropriate research designs, methods, and analytical techniques.

→ Demonstrate Contribution

Well-defined objectives articulate how your study will contribute to existing knowledge and address current debates.

→ Evaluate Success

They provide clear benchmarks for evaluating the success and relevance of your research outcomes.

❑ How to Write Effective Research Objectives

01

Start Broad

Begin with a clear, broad research aim derived from your problem statement.

02

Break Down

Deconstruct the aim into 3-5 specific, measurable objectives.

03

Use Action Verbs

Employ clear and precise action verbs like "analyze," "evaluate," or "determine."

04

Apply SMART Criteria

Ensure each objective is **S**pecific, **M**easurable, **A**chievable, **R**elevant, and **T**ime-bound.

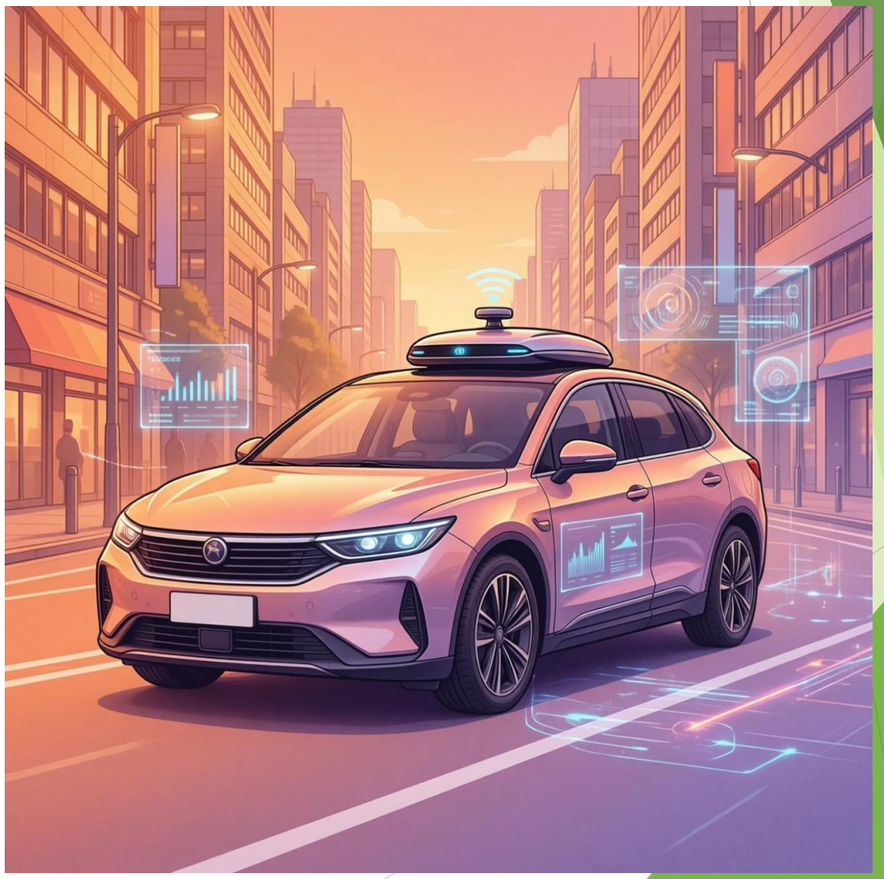
❑ Example of SMART Research Objectives

Aim: To assess safety features and response times of self-driving cars in varied environmental conditions.

S Specific	M Measurable
A Achievable	R Relevant
T Time-bound	

Objectives:

- Measure response time to pedestrians under varied conditions within 6 months.
- Evaluate effectiveness of emergency braking systems in urban settings.
- Analyze driver override frequency during automated driving in highway scenarios.



❑ Summary & Takeaway

Research Objectives: Your Roadmap

They are the essential guide for your study's journey, providing direction and purpose.

Clarity for Impact

Clear objectives ensure your research is focused, efficient, and ultimately impactful.

SMART for Success

Crafting SMART objectives transforms broad aims into precise, actionable goals.

Start Strong

Begin with well-defined objectives to maximize the success and relevance of your research.

END